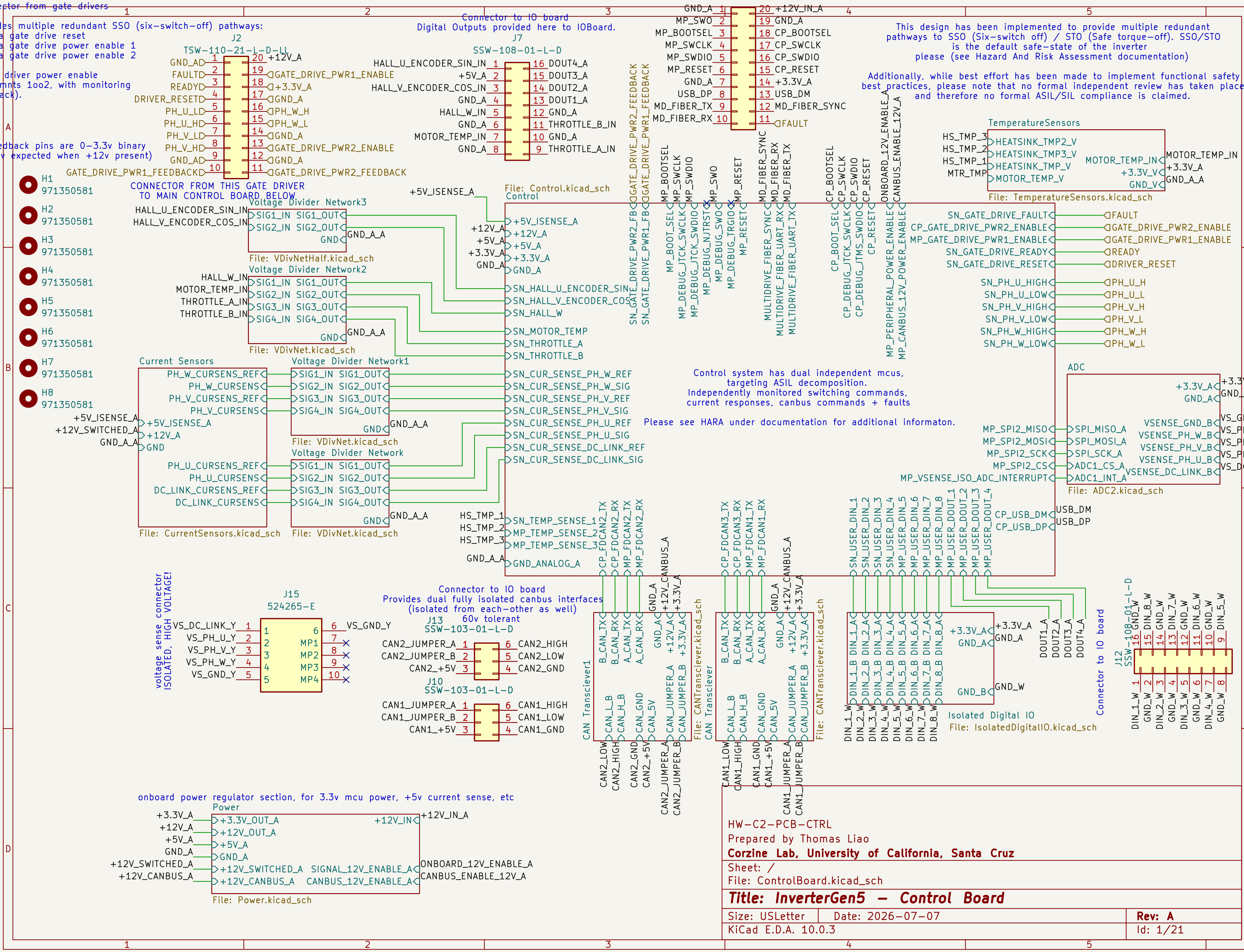
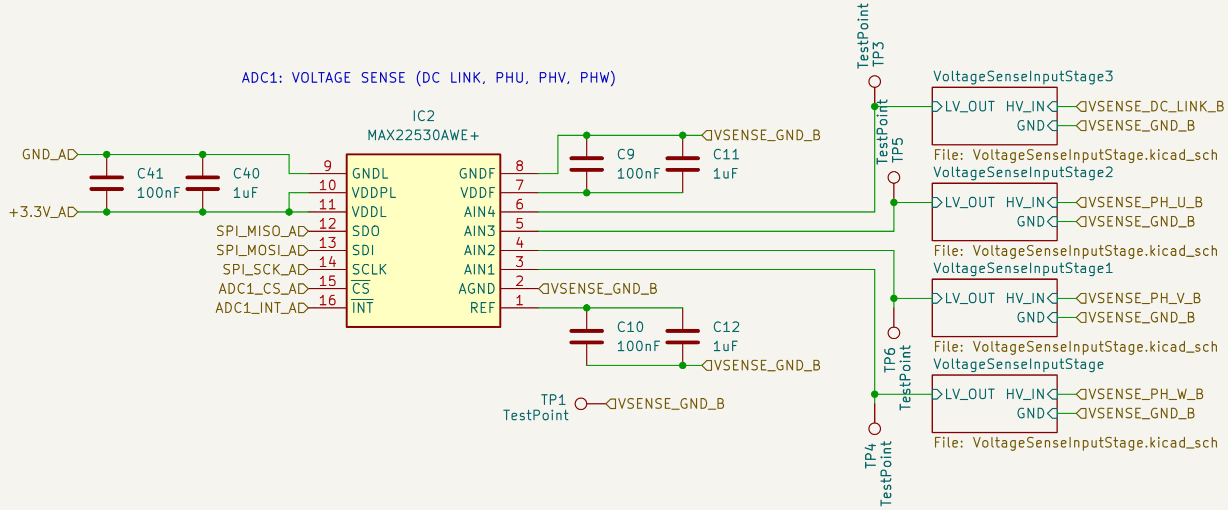

SINGLE WIRE DEBUG / PROGRAMMING
USB INTERFACE TO TOP IO BOARD
+12V TO CTRL BOARD FROM IO BOARD
J14
SSW-110-01-F-D
NOTE: WHEN NOT IN FIBER MULTIDRIVE MULTIPHASE OR LOAD SHARING OPERATION,
DISABLE TIM1 SYNC PIN (TIM1 ETR)
Connector from gate drivers
Includes multiple redundant SSO (six-switch-off) pathways:
- via gate drive reset
- via gate drive power enable 1
- via gate drive power enable 2
(Gate driver power enable implemnts 1oo2, with monitoring feedback).
feedback pins are 0-3.3v binary
(3.3v expected when +12v present)
J2
TSW-110-21-L-D-LL
CONNECTOR FROM THIS GATE DRIVER
TO MAIN CONTROL BOARD BELOW
Connector to IO board
Digital Outputs provided here to IOBoard.
J7
SSW-108-01-L-D
This design has been implemented to provide multiple redundant pathways to SSO (Six-switch off) / STO (Safe torque-off). SSO/STO is the default safe-state of the inverter please (see Hazard And Risk Assessment documentation)
Additionally, while best effort has been made to implement functional safety best practices, please note that no formal independent review has taken place, and therefore no formal ASIL/SIL compliance is claimed.
TemperatureSensors
File: TemperatureSensors.kicad_sch
H1 971350581
H2 971350581
H3 971350581
H4 971350581
H5 971350581
H6 971350581
H7 971350581
H8 971350581
Voltage Divider Network3
File: VDivNetHalf.kicad_sch
Voltage Divider Network2
File: VDivNet.kicad_sch
Current Sensors
File: CurrentSensors.kicad_sch
Voltage Divider Network1
File: VDivNet.kicad_sch
Voltage Divider Network
File: VDivNet.kicad_sch
Control
File: Control.kicad_sch
Control system has dual independent mcus, targeting ASIL decomposition. Independently monitored switching commands, current responses, canbus commands + faults
Please see HARA under documentation for additional informaton.
ADC
File: ADC2.kicad_sch
voltage sense connector
ISOLATED, HIGH VOLTAGE!
J15
524265-E
Connector to IO board
Provides dual fully isolated canbus interfaces
(isolated from each-other as well)
60v tolerant
J13
SSW-103-01-L-D
J10
SSW-103-01-L-D
CAN Transciever1
File: CANTransciever.kicad_sch
CAN Transciever
File: CANTransciever.kicad_sch
Isolated Digital IO
File: IsolatedDigitalIO.kicad_sch
Connector to IO board
J12
SSW-108-01-L-D
onboard power regulator section, for 3.3v mcu power, +5v current sense, etc
Power
File: Power.kicad_sch
HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /
File: ControlBoard.kicad_sch
Title: InverterGen5 - Control Board
Size: USLetter
Date: 2026-07-07
KiCad E.D.A. 10.0.3
Rev: A
Id: 1/21

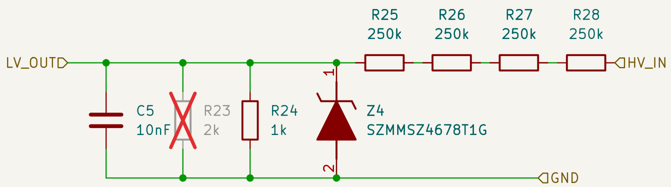


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/
File: ADC2.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 4/21

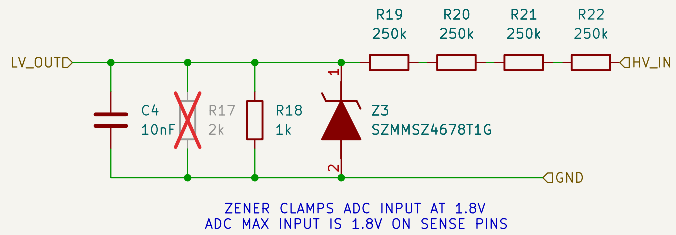


ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /ADC/VoltageSenseInputStage/
File: VoltageSenseInputStage.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 2/21

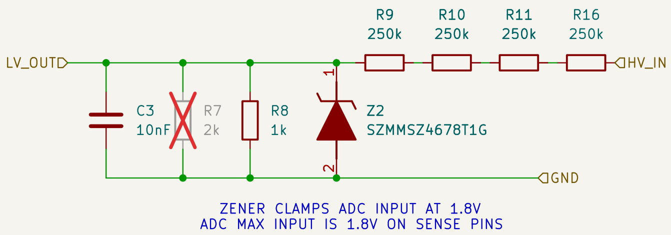


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage1/
File: VoltageSenseInputStage.kicad_sch

Title: InverterGen5 – Control Board

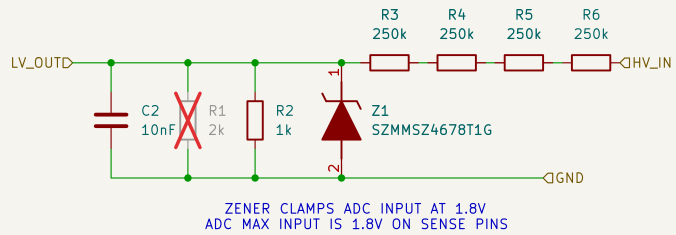
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 3/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /ADC/VoltageSenseInputStage2/
File: VoltageSenseInputStage.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 5/21

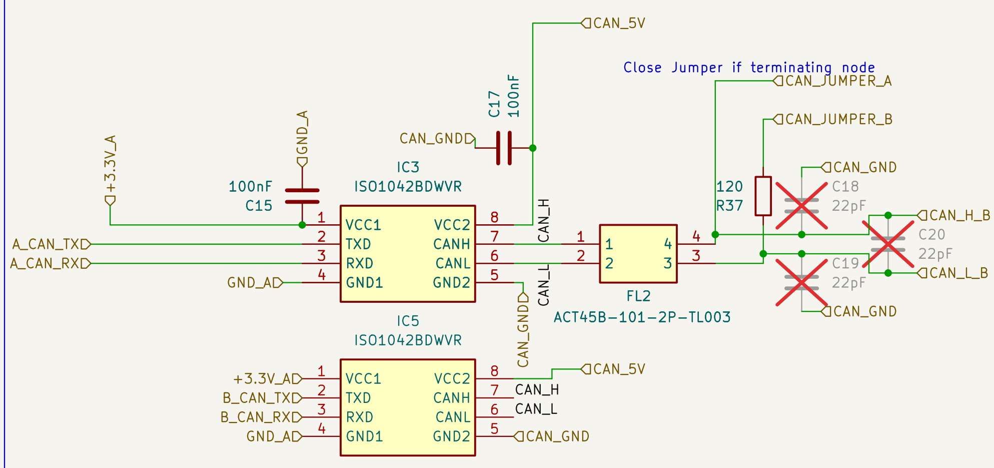


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage3/
File: VoltageSenseInputStage.kicad_sch

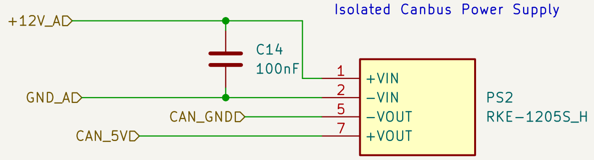
Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 6/21



two can transcievers, allows safety coproc to snoop and interact with can traffic, if needed.
provides redundant path to read and cross-check motor torque commands, if using canbus for torque cmd over throttle A/B inpts

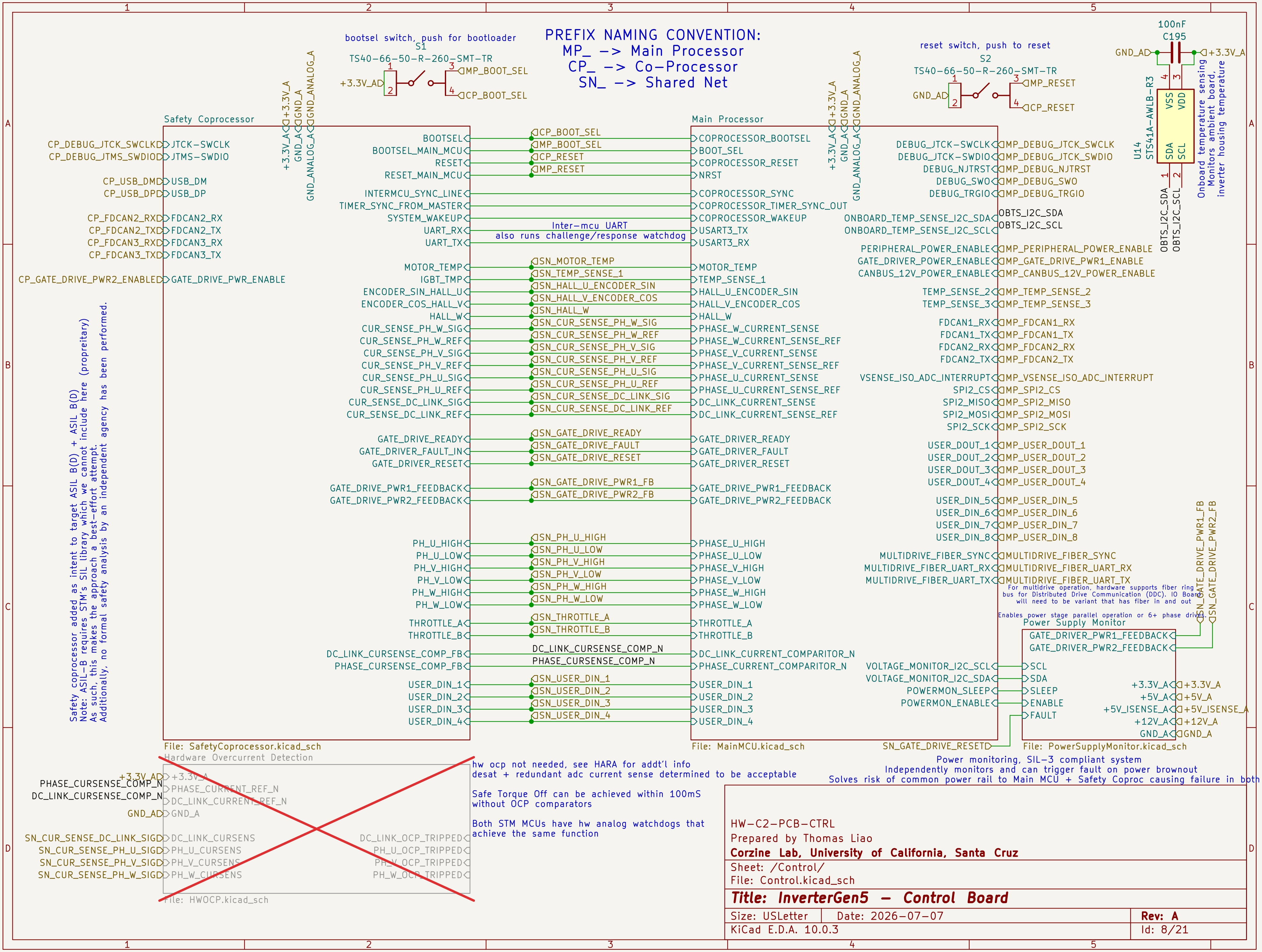
Isolated Canbus Power Supply

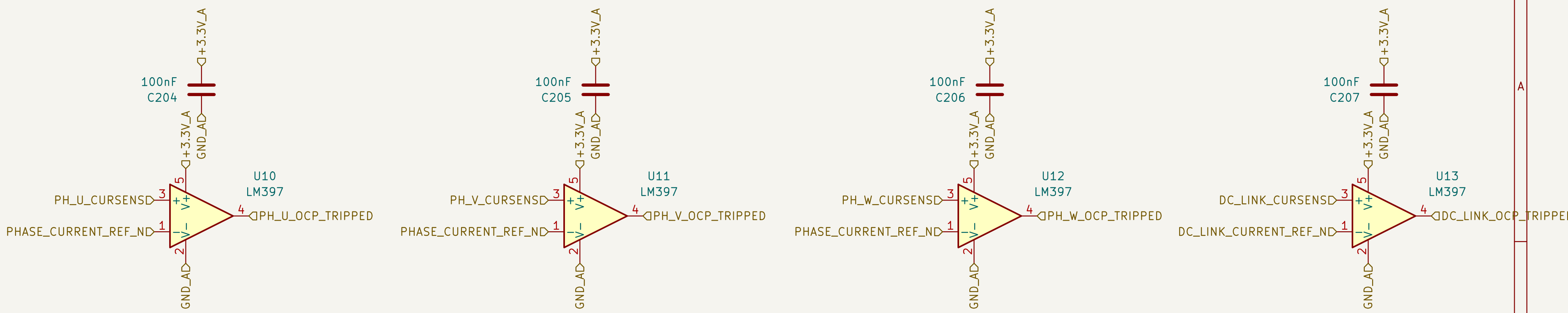


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /CAN Transciever1/
File: CANTransciever.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 7/21


PREFIX NAMING CONVENTION:
MP_ -> Main Processor
CP_ -> Co-Processor
SN_ -> Shared Net
bootsel switch, push for bootloader
S1
TS40-66-50-R-260-SMT-TR
+3.3V_AD
MP_BOOT_SEL
CP_BOOT_SEL
reset switch, push to reset
S2
TS40-66-50-R-260-SMT-TR
GND_AD
MP_RESET
CP_RESET
100nF
C195
GND_AD
+3.3V_A
U14
STS41A-AWLB-R3
VSS
VDD
SDA
SCL
OBTS_I2C_SDA
OBTS_I2C_SCL
Onboard temperature sensing
Monitors ambient board,
inverter housing temperature
Safety Coprocessor
+3.3V_A
GND_A
GND_ANALOG_A
CP_DEBUG_JTCK_SWCLK
JTCK-SWCLK
CP_DEBUG_JTMS_SWDIO
JTMS-SWDIO
CP_USB_DM
USB_DM
CP_USB_DP
USB_DP
CP_FDCAN2_RX
FDCAN2_RX
CP_FDCAN2_TX
FDCAN2_TX
CP_FDCAN3_RX
FDCAN3_RX
CP_FDCAN3_TX
FDCAN3_TX
CP_GATE_DRIVE_PWR2_ENABLE
GATE_DRIVE_PWR_ENABLE
BOOTSEL
BOOTSEL_MAIN_MCU
RESET
RESET_MAIN_MCU
INTERMCU_SYNC_LINE
TIMER_SYNC_FROM_MASTER
SYSTEM_WAKEUP
UART_RX
UART_TX
MOTOR_TEMP
IGBT_TMP
ENCODER_SIN_HALL_U
ENCODER_COS_HALL_V
HALL_W
CUR_SENSE_PH_W_SIG
CUR_SENSE_PH_W_REF
CUR_SENSE_PH_V_SIG
CUR_SENSE_PH_V_REF
CUR_SENSE_PH_U_SIG
CUR_SENSE_PH_U_REF
CUR_SENSE_DC_LINK_SIG
CUR_SENSE_DC_LINK_REF
GATE_DRIVE_READY
GATE_DRIVER_FAULT_IN
GATE_DRIVER_RESET
GATE_DRIVE_PWR1_FEEDBACK
GATE_DRIVE_PWR2_FEEDBACK
PH_U_HIGH
PH_U_LOW
PH_V_HIGH
PH_V_LOW
PH_W_HIGH
PH_W_LOW
THROTTLE_A
THROTTLE_B
DC_LINK_CURSENSE_COMP_FB
PHASE_CURSENSE_COMP_FB
USER_DIN_1
USER_DIN_2
USER_DIN_3
USER_DIN_4
File: SafetyCoprocessor.kicad_sch
CP_BOOT_SEL
MP_BOOT_SEL
CP_RESET
MP_RESET
Inter-mcu UART
also runs challenge/response watchdog
SN_MOTOR_TEMP
SN_TEMP_SENSE_1
SN_HALL_U_ENCODER_SIN
SN_HALL_V_ENCODER_COS
SN_HALL_W
SN_CUR_SENSE_PH_W_SIG
SN_CUR_SENSE_PH_W_REF
SN_CUR_SENSE_PH_V_SIG
SN_CUR_SENSE_PH_V_REF
SN_CUR_SENSE_PH_U_SIG
SN_CUR_SENSE_PH_U_REF
SN_CUR_SENSE_DC_LINK_SIG
SN_CUR_SENSE_DC_LINK_REF
SN_GATE_DRIVE_READY
SN_GATE_DRIVE_FAULT
SN_GATE_DRIVE_RESET
SN_GATE_DRIVE_PWR1_FB
SN_GATE_DRIVE_PWR2_FB
SN_PH_U_HIGH
SN_PH_U_LOW
SN_PH_V_HIGH
SN_PH_V_LOW
SN_PH_W_HIGH
SN_PH_W_LOW
SN_THROTTLE_A
SN_THROTTLE_B
DC_LINK_CURSENSE_COMP_N
PHASE_CURSENSE_COMP_N
SN_USER_DIN_1
SN_USER_DIN_2
SN_USER_DIN_3
SN_USER_DIN_4
Main Processor
COPROCESSOR_BOOTSEL
BOOT_SEL
COPROCESSOR_RESET
NRST
COPROCESSOR_SYNC
COPROCESSOR_TIMER_SYNC_OUT
COPROCESSOR_WAKEUP
USART3_TX
USART3_RX
MOTOR_TEMP
TEMP_SENSE_1
HALL_U_ENCODER_SIN
HALL_V_ENCODER_COS
HALL_W
PHASE_W_CURRENT_SENSE
PHASE_W_CURRENT_SENSE_REF
PHASE_V_CURRENT_SENSE
PHASE_V_CURRENT_SENSE_REF
PHASE_U_CURRENT_SENSE
PHASE_U_CURRENT_SENSE_REF
DC_LINK_CURRENT_SENSE
DC_LINK_CURRENT_SENSE_REF
GATE_DRIVER_READY
GATE_DRIVER_FAULT
GATE_DRIVER_RESET
GATE_DRIVE_PWR1_FEEDBACK
GATE_DRIVE_PWR2_FEEDBACK
PHASE_U_HIGH
PHASE_U_LOW
PHASE_V_HIGH
PHASE_V_LOW
PHASE_W_HIGH
PHASE_W_LOW
THROTTLE_A
THROTTLE_B
DC_LINK_CURRENT_COMPARITOR_N
PHASE_CURRENT_COMPARITOR_N
USER_DIN_1
USER_DIN_2
USER_DIN_3
USER_DIN_4
DEBUG_JTCK-SWCLK
MP_DEBUG_JTCK_SWCLK
DEBUG_JTCK-SWDIO
MP_DEBUG_JTCK_SWDIO
DEBUG_NJTRST
MP_DEBUG_NJTRST
DEBUG_SWO
MP_DEBUG_SWO
DEBUG_TRGIO
MP_DEBUG_TRGIO
ONBOARD_TEMP_SENSE_I2C_SDA
OBTS_I2C_SDA
ONBOARD_TEMP_SENSE_I2C_SCL
OBTS_I2C_SCL
PERIPHERAL_POWER_ENABLE
MP_PERIPHERAL_POWER_ENABLE
GATE_DRIVER_POWER_ENABLE
MP_GATE_DRIVE_PWR1_ENABLE
CANBUS_12V_POWER_ENABLE
MP_CANBUS_12V_POWER_ENABLE
TEMP_SENSE_2
MP_TEMP_SENSE_2
TEMP_SENSE_3
MP_TEMP_SENSE_3
FDCAN1_RX
MP_FDCAN1_RX
FDCAN1_TX
MP_FDCAN1_TX
FDCAN2_RX
MP_FDCAN2_RX
FDCAN2_TX
MP_FDCAN2_TX
VSENSE_ISO_ADC_INTERRUPT
MP_VSENSE_ISO_ADC_INTERRUPT
SPI2_CS
MP_SPI2_CS
SPI2_MISO
MP_SPI2_MISO
SPI2_MOSI
MP_SPI2_MOSI
SPI2_SCK
MP_SPI2_SCK
USER_DOUT_1
MP_USER_DOUT_1
USER_DOUT_2
MP_USER_DOUT_2
USER_DOUT_3
MP_USER_DOUT_3
USER_DOUT_4
MP_USER_DOUT_4
USER_DIN_5
MP_USER_DIN_5
USER_DIN_6
MP_USER_DIN_6
USER_DIN_7
MP_USER_DIN_7
USER_DIN_8
MP_USER_DIN_8
MULTIDRIVE_FIBER_SYNC
MULTIDRIVE_FIBER_UART_RX
MULTIDRIVE_FIBER_UART_TX
For multidrive operation, hardware supports fiber ring bus for Distributed Drive Communication (DDC). IO Board will need to be variant that has fiber in and out
Enables power stage parallel operation or 6+ phase drives
VOLTAGE_MONITOR_I2C_SCL
VOLTAGE_MONITOR_I2C_SDA
POWERMON_SLEEP
POWERMON_ENABLE
SN_GATE_DRIVE_RESET
File: MainMCU.kicad_sch
Power Supply Monitor
GATE_DRIVER_PWR1_FEEDBACK
GATE_DRIVER_PWR2_FEEDBACK
SN_GATE_DRIVE_PWR1_FB
SN_GATE_DRIVE_PWR2_FB
SCL
SDA
SLEEP
ENABLE
FAULT
+3.3V_A
+5V_A
+5V_ISENSE_A
+12V_A
GND_A
File: PowerSupplyMonitor.kicad_sch
Power monitoring, SIL-3 compliant system
Independently monitors and can trigger fault on power brownout
Solves risk of common power rail to Main MCU + Safety Coproc causing failure in both
Safety coprocessor added as intent to target ASIL B(D) + ASIL B(D)
Note: ASIL-B requires STM's SIL library which we cannot include here (proprietary)
As such, this makes the approach a best-effort attempt.
Additionally, no formal safety analysis by an independent agency has been performed.
Hardware Overcurrent Detection
+3.3V_AD
+3.3V_A
PHASE_CURSENSE_COMP_N
PHASE_CURRENT_REF_N
DC_LINK_CURSENSE_COMP_N
DC_LINK_CURRENT_REF_N
GND_AD
GND_A
SN_CUR_SENSE_DC_LINK_SIG
DC_LINK_CURSENS
SN_CUR_SENSE_PH_U_SIG
PH_U_CURSENS
SN_CUR_SENSE_PH_V_SIG
PH_V_CURSENS
SN_CUR_SENSE_PH_W_SIG
PH_W_CURSENS
DC_LINK_OCP_TRIPPED
PH_U_OCP_TRIPPED
PH_V_OCP_TRIPPED
PH_W_OCP_TRIPPED
File: HWOCP.kicad_sch
hw ocp not needed, see HARA for addt'l info
desat + redundant adc current sense determined to be acceptable
Safe Torque Off can be achieved within 100mS
without OCP comparators
Both STM MCUs have hw analog watchdogs that
achieve the same function
HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Control/
File: Control.kicad_sch
Title: InverterGen5 - Control Board
Size: USLetter
Date: 2026-07-07
Rev: A
KiCad E.D.A. 10.0.3
Id: 8/21

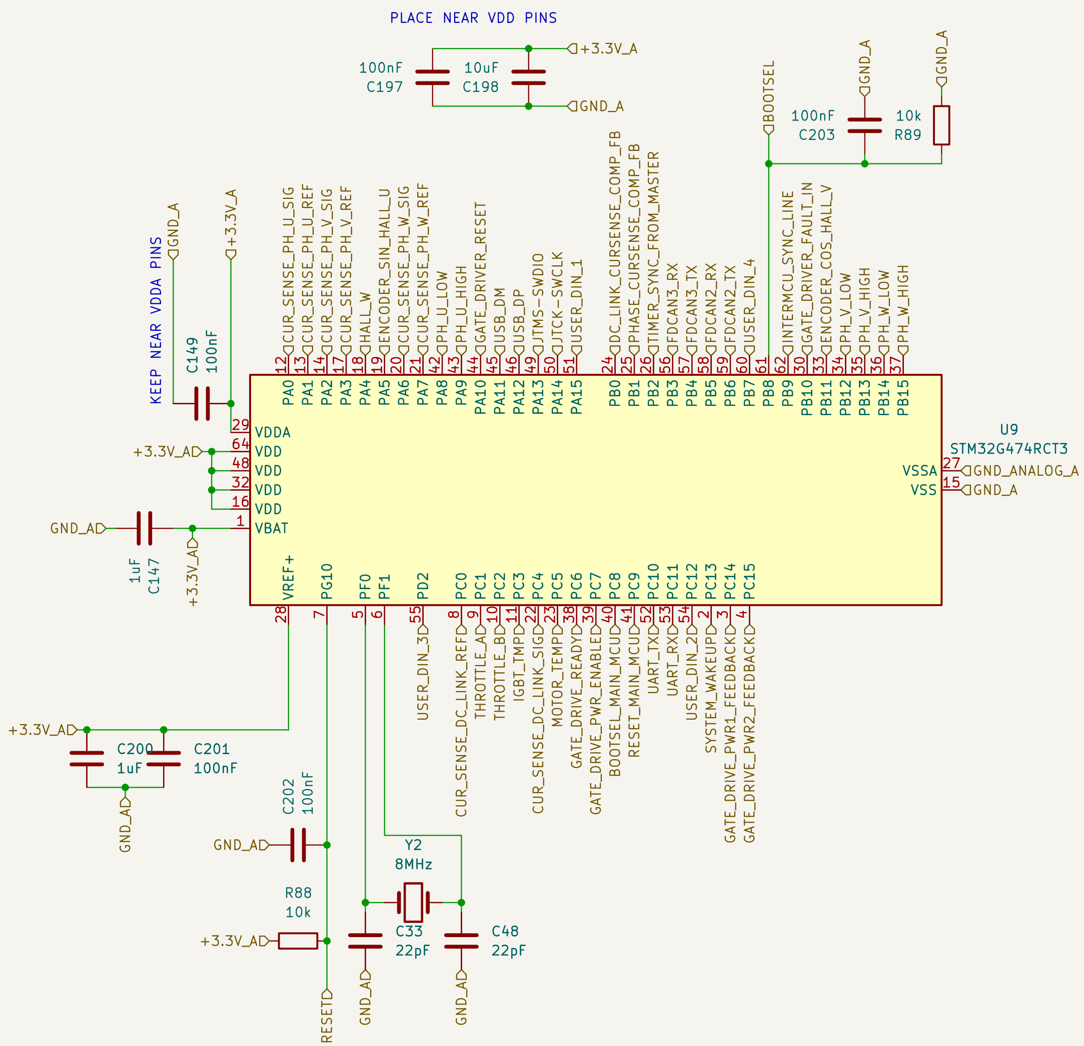


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Control/Hardware Overcurrent Detection/
File: HWOCP.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 10/21



HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

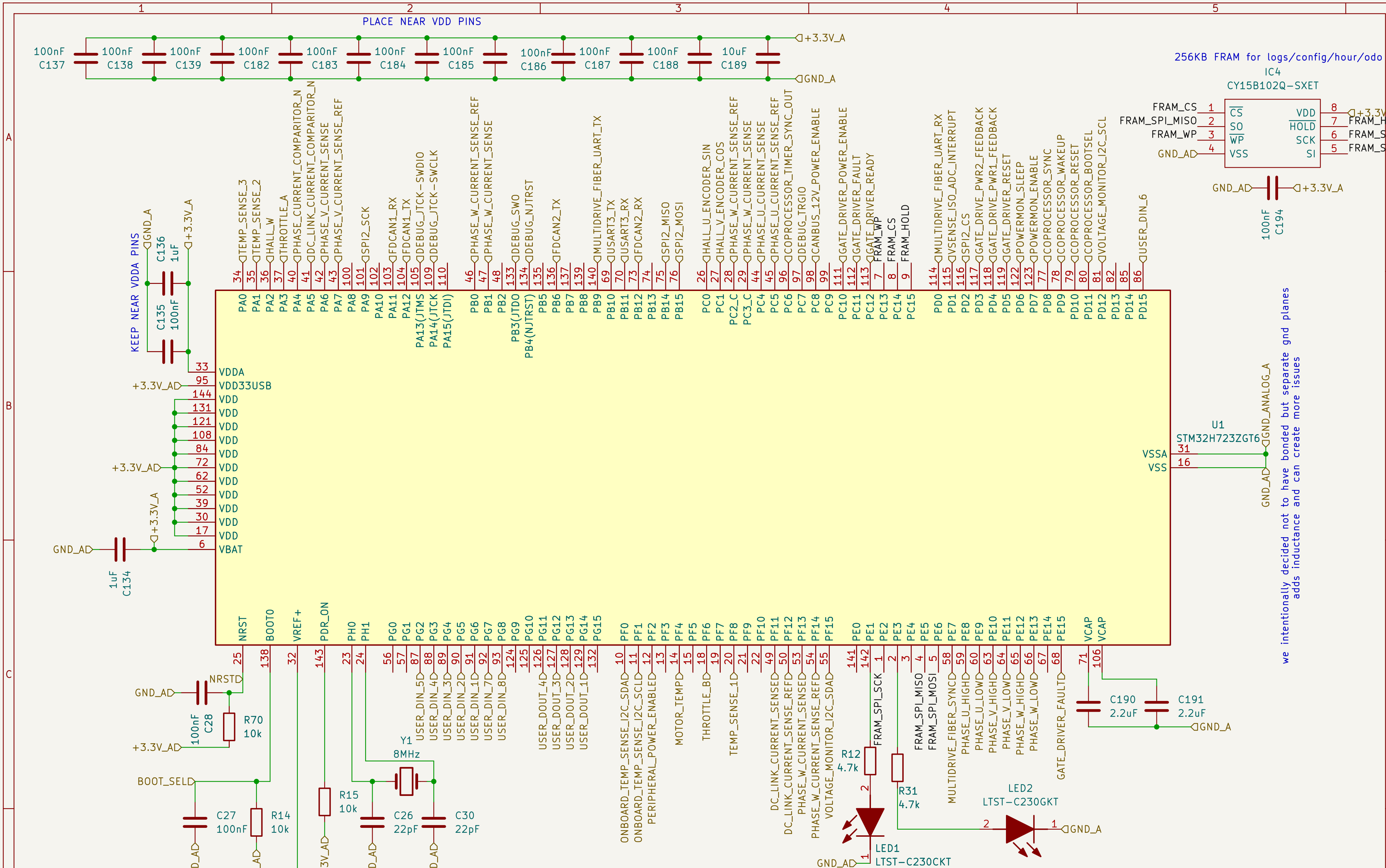
Sheet: /Control/Safety Coprocessor/

File: SafetyCoprocessor.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter | Date: 2026-07-07 | **Rev: A**

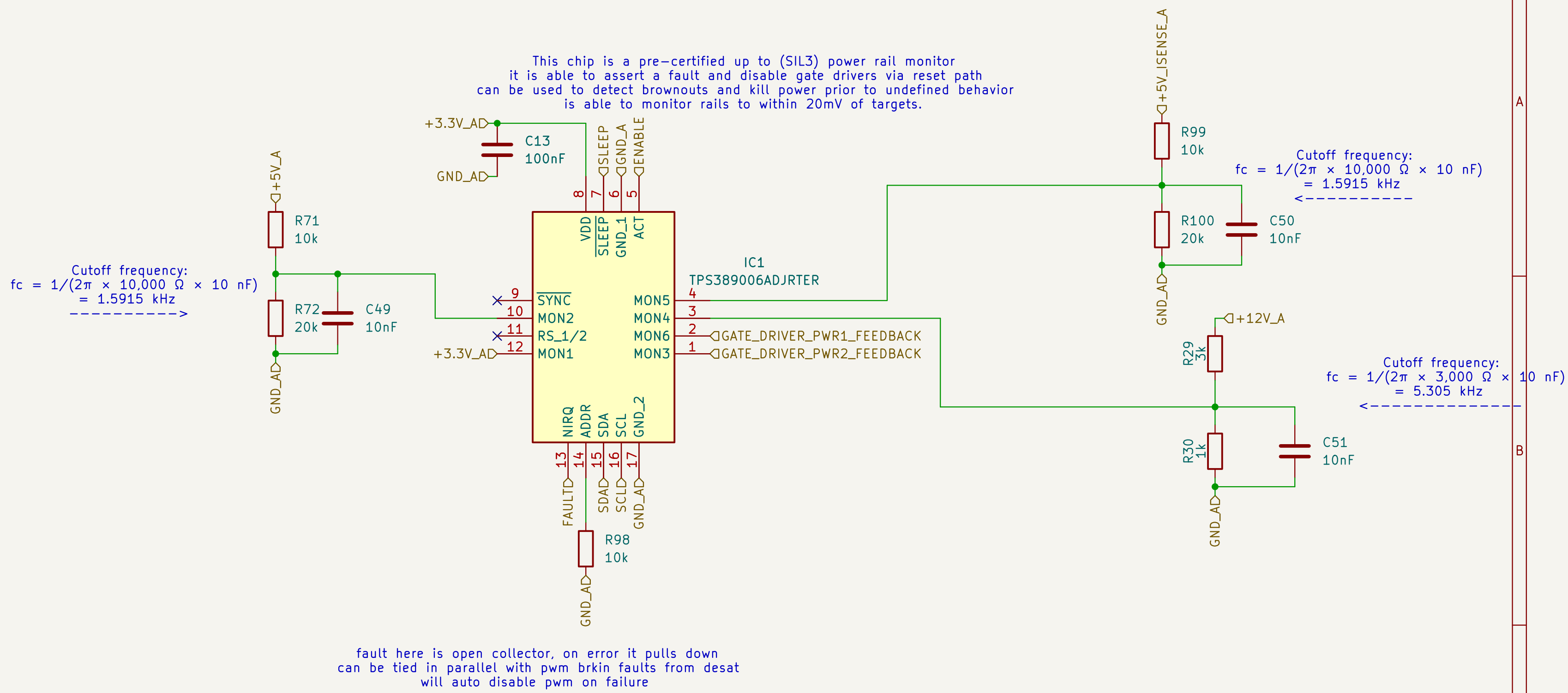
KiCad E.D.A. 10.0.3 | Id: 16/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Control/Main Processor/
File: MainMCU.kicad_sch

Title: InverterGen5 – Control Board

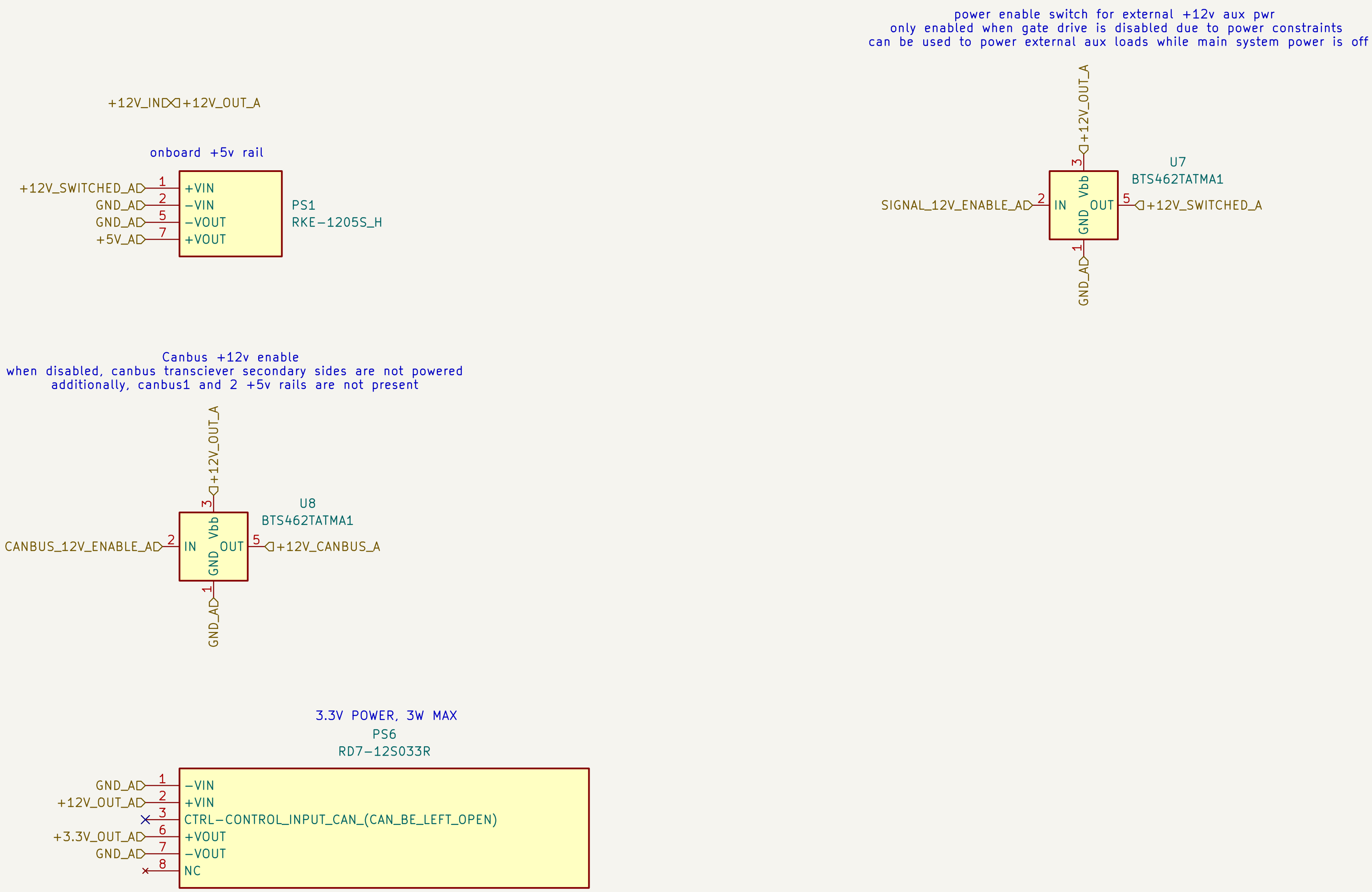
Size: USLetter	Date: 2026-07-07	Rev: A
KiCad E.D.A. 10.0.3		Id: 19/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Control/Power Supply Monitor/
File: PowerSupplyMonitor.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter | Date: 2026-07-07 | **Rev: A**
KiCad E.D.A. 10.0.3 | Id: 20/21

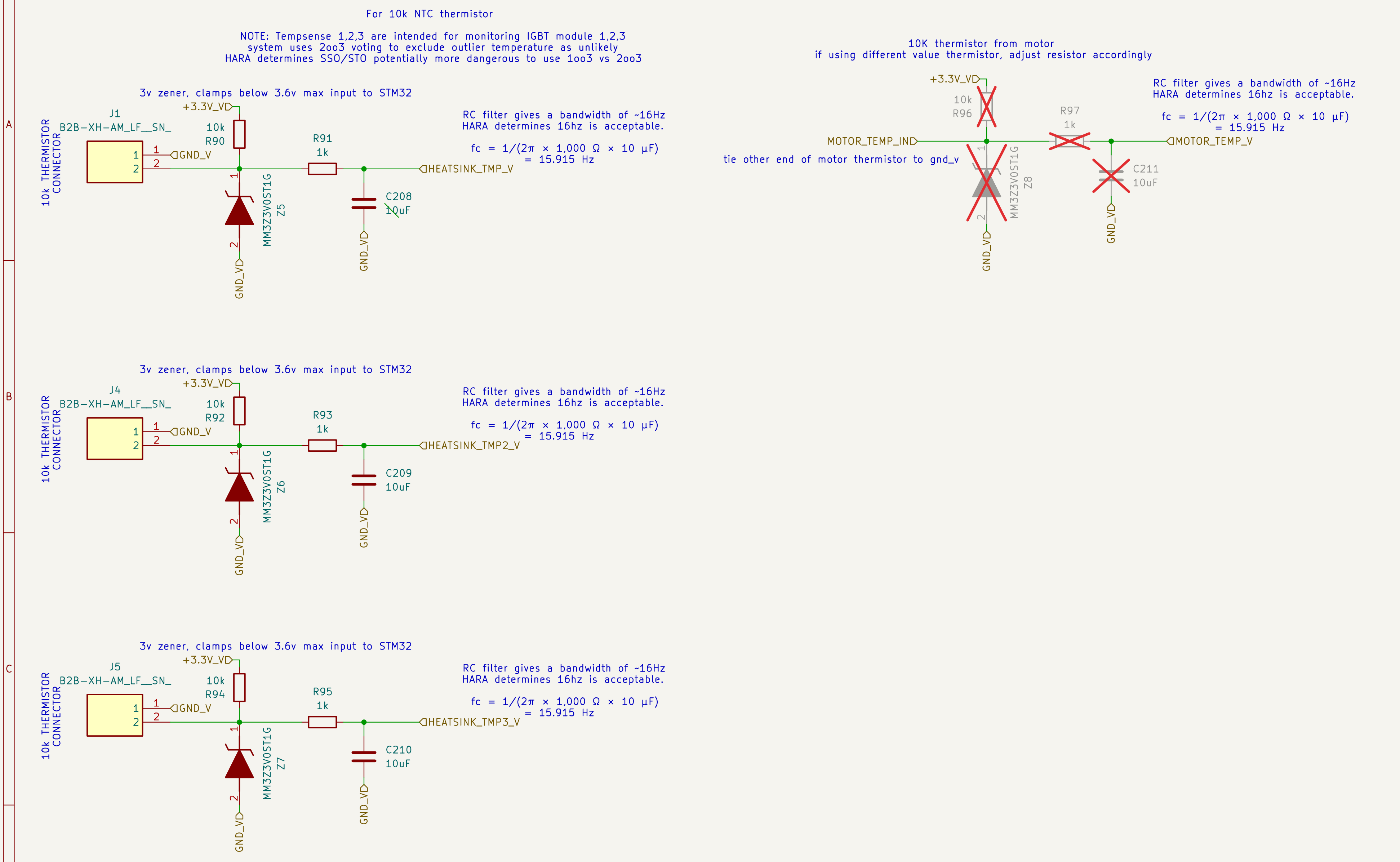


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Power/
File: Power.kicad_sch

Title: InverterGen5 - Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 9/21



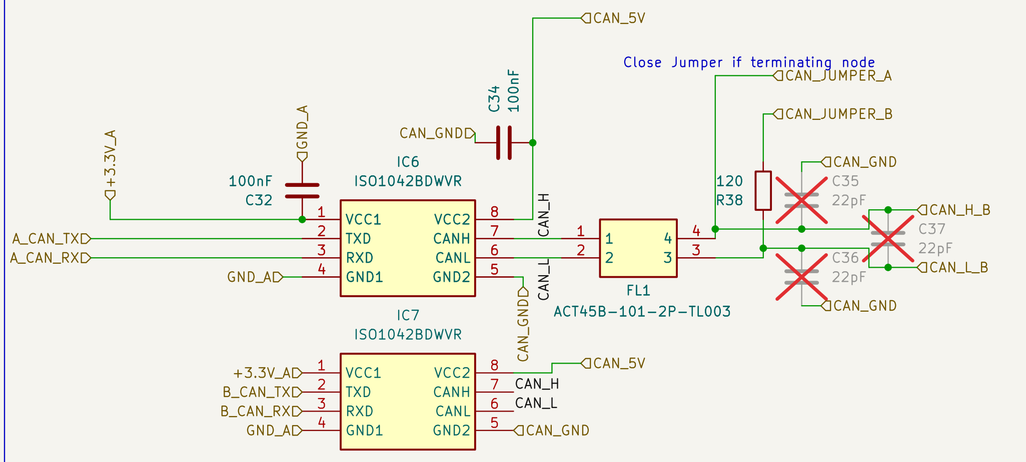
HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /TemperatureSensors/
File: TemperatureSensors.kicad_sch

Title: InverterGen5 – Control Board

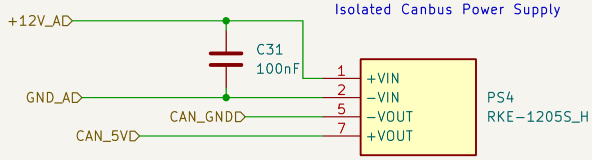
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 9/21

Close Jumper if terminating node



two can transcievers, allows safety coproc to snoop and interact with can traffic, if needed.
provides redundant path to read and cross-check motor torque commands, if using canbus for torque cmd over throttle A/B inpts

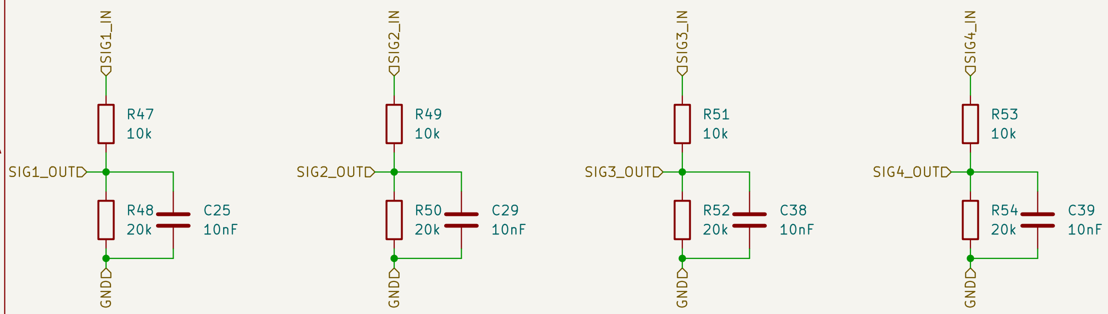
Isolated Canbus Power Supply



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /CAN Transciever/
File: CANTransciever.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 11/21



HW-C2-PCB-CTRL

Prepared by Thomas Liao

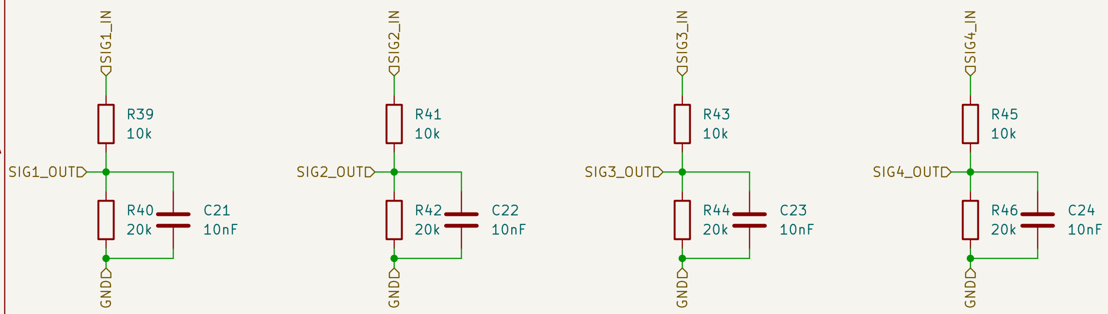
Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network1/

File: VDivNet.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 12/21

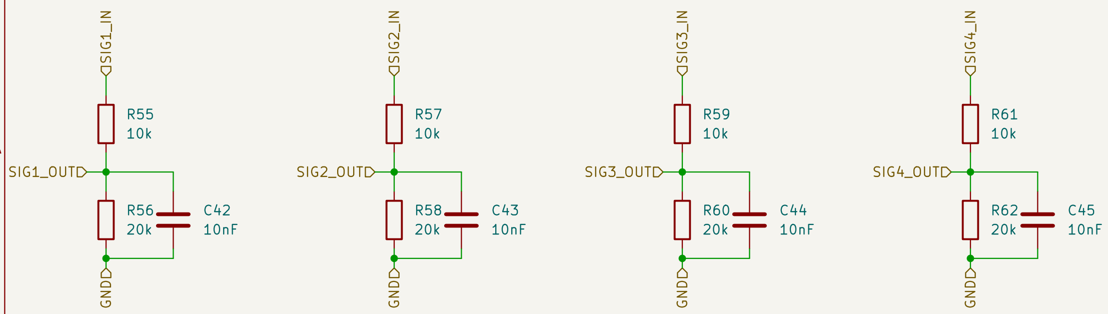


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network/
File: VDivNet.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 13/21

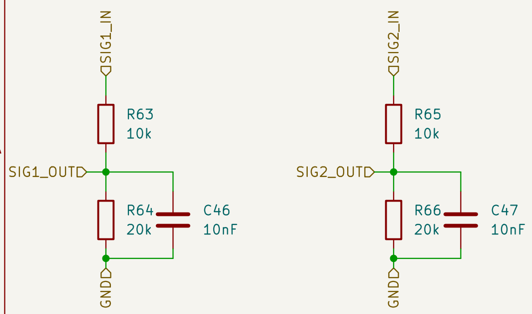


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network2/
File: VDivNet.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 14/21



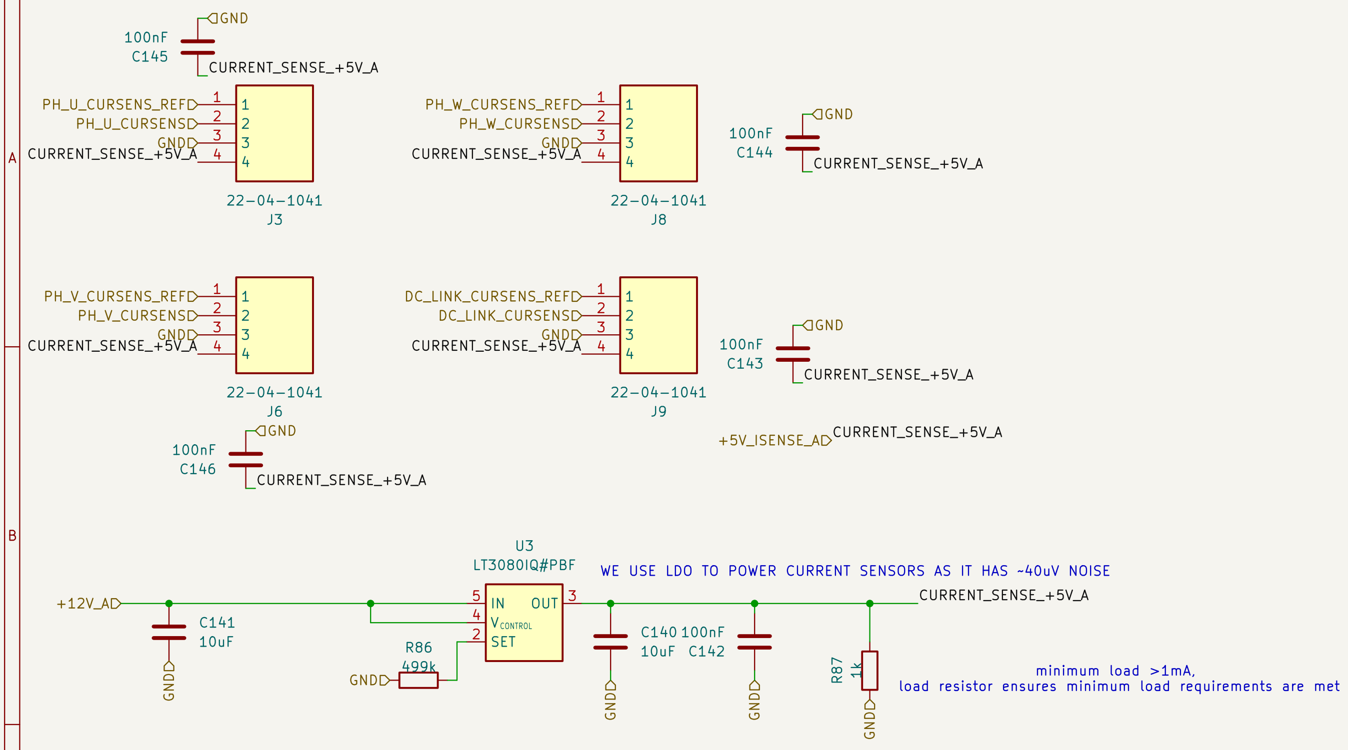
HW-C2-PCB-CTRL

Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network3/
File: VDivNetHalf.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 15/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
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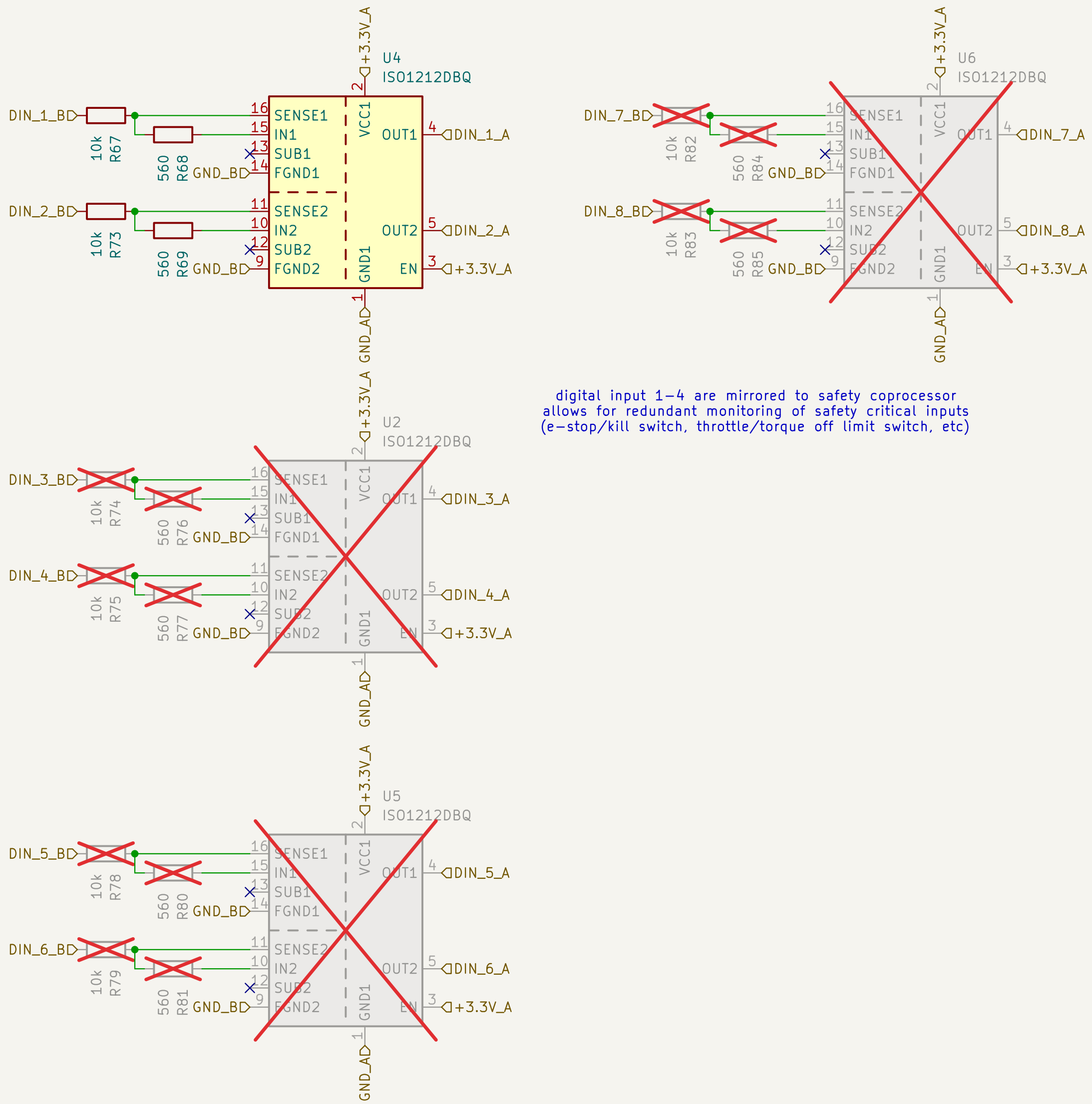
Sheet: /Current Sensors/
File: CurrentSensors.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 17/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
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Sheet: /Isolated Digital IO/
File: IsolatedDigitalIO.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
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